Blow-up behavior for reaction-diffusion equations with non scale invariant nonlinearity

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Equation

We consider

$$\begin{cases} u_t - \Delta u = f(u), & x \in \Omega,\ t > 0, \\ u = 0, & x \in \partial\Omega,\ t > 0, , \\ u(x,0) = u_0(x), & x \in \Omega, \end{cases} \tag{1}$$

where $\Omega \subset \mathbb{R}^n$ uniformly smooth domain ($n \geq 1$). We define u as classical blow-up solution of (1) defined on $\Omega \times (0,T)$.

Assumptions

We consider the following subclass of functions with regular variation, with index $p > 1$ (i.e $f(s) = s^p L(s)$):

$$f \in C^1([0,\infty)), \quad f(0) \geq 0, \quad f \text{ is positive and } C^2 \text{ for large } s \tag{2}$$

and

$$\frac{sL'(s)}{L(s)} = O\big(\log^{-\alpha}(s)\big) \text{ as } s \to \infty, \text{ for some } \alpha > \tfrac{1}{2}. \tag{3}$$

Define

$$C_0(\mathbb{R}^n) := \{\varphi \in C(\mathbb{R}^n);\ \lim_{|x|\to\infty} \varphi(x) = 0\} \quad \text{and} \quad G(X) := \int_X^\infty \frac{ds}{f(s)}.$$

Assuming $p < p_s$, all solution of (1) is of type I, cf. [5] and [6] fore more general cases.

Examples

- As examples of nonlinearities such that assumptions (2)-(3) are satisfied, we have $f(s) = s^p L(s)$ with L given by:
  - $\star$ $\log^a(K+s)$ for $K > 1$ and $a \in \mathbb{R}$,
  - $\star$ the iterated logarithms $\log_m(K+s)$,
  - $\star$ $\exp\left(|\log s|^\nu\right)$ with $\nu \in (0, 1/2)$,
  - $\star$ $\big[\log(3+s)\big]^{\sin[\log\log(3+s)]}$
  - $\star$ $1 + a\sin\big(\log^\nu(2+s)\big)$ with $\nu \in (0,1/2)$ and $|a| < 1$.
- Example of stable type I blowup solution was constructed in [4] for the logarithmic nonlinearities cases.

Convergence: Giga-Kohn type property ([1])

Let $1 < p < p_S$ and $u_0 \in L^\infty(\Omega)$ with $u_0 \geq 0$. Assume (2)-(3) and $T := T_{\max}(u_0) < \infty$. If $a \in \Omega$ is a blow-up point of u, then

$$\lim_{t\to T} \frac{u(a + y\sqrt{T-t}, t)}{G^{-1}(T-t)} = 1,$$

uniformly on compact sets $|y| \leq C$.

Compactness of blow-up set ([1])

Let $1 < p < p_S$, assume (2)-(3), $f(0) = 0$, and let $u_0 \in C_0(\mathbb{R}^n)$ with $u_0 \geq 0$ be such that $T < \infty$. Then the blow-up set of u is compact. More precisely, there exists $R > 0$ such that

$$\sup_{|x|>R,\, t\in(0,T)} u < \infty.$$

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Blow-up profiles ([3])

Let $1 < p < p_S$, $\Omega = \mathbb{R}^n$ or $\Omega = B_R$, and assume (2)-(3) and that

$$\left(\frac{sL'}{L}\right)' = o\left(\frac{1}{s\log s}\right) \text{ as } s \to \infty.$$

Let $u_0 \geq 0$ be radially symmetric, nonincreasing in $|x|$, and assume $T < \infty$. If $\Omega = \mathbb{R}^n$ assume also that u_0 is nonconstant. Then we have:

(i) **(blow-up profile in original variables)**

$$u(x,t) = (1+o(1))G^{-1}\left(T - t + \frac{p-1}{4p}\frac{|x|^2}{|\log|x|^2|}\right) \quad \text{as } (x,t) \to (0,T). \tag{4}$$

(ii) **(final profile)**

$$u(x,T) = (1+o(1))G^{-1}\left(\frac{(p-1)|x|^2}{8p|\log|x||}\right) \quad \text{as } x \to 0;$$

(iii) **(refined space-time profile)**

$$u\big(\xi\sqrt{(T-t)|\log(T-t)|}, t\big) = (1+o(1))G^{-1}\left[(T-t)\left(1 + \frac{(p-1)|\xi|^2}{4p}\right)\right]$$

as $t \to T$ and uniformly for ξ bounded.

Structural universality of the profile

As a remarkable fact and completely new observation, we see that the global blow-up profile (4) possesses a universal structure, which is to a large extent independent of the nonlinearity. Namely it is given by the "resolvent" G^{-1} of the ODE, composed with a universal, time-space building block

$$T - t + \frac{p-1}{4p}\frac{|x|^2}{|\log|x|^2|}$$

which is the same as in the pure power case. In particular, the building block of the profile is not affected by the slowly varying factor L of the nonlinearity and, at principal order, the effect of the latter on the profile is reflected only in the application of the resolvent G^{-1} to this building block.

Exponential nonlinearities ([2])

Assuming u_0, Ω and L as before, the similar results, convergence, blow-up profile and so structural universality of the profile, hold in the case $f(u) = e^u L(e^u)$.

References

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